

## EXPERIMENT 9:

### 9. Illustrate the concept of inter-process communication using shared memory with a C program

Online C Compiler

Share

Run

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <sys/ipc.h>
6 #include <sys/shm.h>
7
8 #define SHM_SIZE 1024
9
10 int main() {
11     key_t key = ftok("shmfile", 65);
12
13     int shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666);
14     if (shmid == -1) {
15         perror("shmget");
16         exit(EXIT_FAILURE);
17     }
18
19     char *shm_ptr = (char*)shmat(shmid, NULL, 0);
20     if (shm_ptr == (char*)(-1)) {
21         perror("shmat");
22         exit(EXIT_FAILURE);
23     }
24
25     strcpy(shm_ptr, "Hello, shared memory!");
26
27     printf("Data written to shared memory: %s\n", shm_ptr);
28
29     if (shmdt(shm_ptr) == -1) {
30         perror("shmdt");
31         exit(EXIT_FAILURE);
32     }
33
34     if (shmctl(shmid, IPC_RMID, NULL) == -1) {
35         perror("shmctl");
36         exit(EXIT_FAILURE);
37     }
38
39     return 0;
40 }
```

Output

Status: **Accepted**

Memory Used: 756 byte | Execution Time: 0.003 sec

Data written to shared memory: Hello, shared memory!

Input

Enter your input here...